
TROPIC01

ODR_TR01_PSR_006_FI_KUL

TROPIC01 Fault Injection Evaluation

Version: 1.0

Abstract

This report presents a fault injection evaluation of the TROPIC01 secure element, performed by the COSIC research group at KU Leuven. COSIC assessed the feasibility of voltage, electromagnetic, and laser fault injection attacks and the effectiveness of the integrated security sensors. The evaluation produced several findings, all of which Tropic Square has addressed. Some are partially addressed in the current silicon revision through a firmware update and others will be addressed in a future silicon revision.

June 23, 2026





Summary of the Security Report

Report Details

- **Evaluation Laboratory:** KU Leuven, COSIC
- **Evaluation Laboratory Contact:** Lennert Wouters lennert.wouters@esat.kuleuven.be,
Benedikt Gierlichs benedikt.gierlichs@esat.kuleuven.be
- **Contract Type:** Paid Contract
- **Evaluation Period:** 03/2025 - 05/2026
- **Type of Evaluation:** Open Samples: custom samples prepared by Tropic Square
- **Tropic Square Contact:** Tropic Square Security Team,
security@tropicsquare.com

Tropic Square Executive Summary

At the request of Tropic Square, the COSIC research group at KU Leuven performed a scope-limited evaluation of the TROPIC01 secure element, assessing the feasibility of fault injection attacks and the effectiveness of its integrated security sensors. The evaluation covered voltage, electromagnetic, and laser fault injection, and validated the supply voltage monitor, voltage glitch, EM pulse, and laser detectors.

COSIC found the sensors to be generally active and effective at detecting the fault injection mechanisms they target. However, the supply voltage monitor was not enabled after power-up due to a ROM bootloader bug, and in several cases faults could be injected without triggering any sensor. Building on these results, COSIC demonstrated a successful laser fault injection attack against the SPECT EdDSA signature verification, which protects secure boot and firmware updates [4].

Tropic Square has addressed the findings on two fronts. A firmware update (Mutable FW v0.3.1) has been released that enables the supply voltage monitor at the earliest possible stage of the boot process on the current silicon revision.

In future silicon revisions, the RISC-V and SPECT ROM bootloader code has been hardened against fault injection, following the principle of defense in depth to complement the hardware sensors. In support of this work, a SPECT fault emulation tool has been integrated into Tropic Square's development and validation pipeline. The detailed findings and Tropic Square's response are summarized in the remainder of this report.



Tropic Square Recommendations

Tropic Square regularly publishes FW updates to their products and resolves product weaknesses responsibly, as described in the Coordinated Vulnerability Disclosure Policy (CVD) (<http://tropicsquare.com/cvd>).

Confidentiality Statement

Several confidential reports underline this public report. These confidential reports served for TROPIC01 improvement and are covered by a mutual NDA between Tropic Square and KU Leuven, COSIC.

During evaluations, Tropic Square provided the partner KU Leuven, COSIC with open samples, with modified or deactivated attack countermeasures, or modified firmware providing extended access to the TROPIC01 to allow detailed individual evaluation of measures stacked in the defense-in-depth paradigm.

Because Tropic Square is devoted to protecting the interests of its customers, detailed findings or test vectors are protected under NDA to prevent exploitation of these data by malicious actors. Here, Tropic Square follows the recommendations of ISO 27001, Annex A 6.6. See Disclosure of Findings below.

However, this fact does not affect the openness of Tropic Square: everybody is welcome to perform independent standard sample evaluation, as defined by our Coordinated Vulnerability Disclosure Policy (CVD) (<http://tropicsquare.com/cvd>).

Any disclosure of the weakness or vulnerability measures is not directly contained in this report, but is processed according to our CVD.



Disclosure of Findings

Tropic Square discloses security findings as openly as possible in public security reports. The only exception is when disclosing sensitive information would significantly reduce the cost of exploiting a particular weakness.

If information could aid exploitation, Tropic Square may withhold it if obtaining that information is beyond the reach of an attacker with the capability to exploit the weakness. We assess whether sensitive information is within an attacker's reach by evaluating it across four barriers. If disclosure would cross any barrier, Tropic Square may choose not to disclose the information to protect device security and its customers:

- **Cost** — Equipment cost to obtain the information exceeds the cost-of-ownership of equipment needed for exploitation (including rentals/services).
- **Time** — Time to obtain the information is significant (≥ 1 week).
- **Expertise** — Obtaining the information requires expertise in a different field than exploitation demands (including combined or supply chain attacks).
- **Capability** — Obtaining the information requires a higher adversary capability level than exploiting the weakness (e.g., layman-level knowledge suffices for exploitation, but expert-level is needed to obtain the information).

Disclaimer: These barriers do not apply to the Coordinated Vulnerability Disclosure (CVD) process: information disclosed through CVD is published in full.

For the assessment, we follow a method inspired by the Common Criteria certification assessment, developed in the ORSHIN research project. [3]

Version History

Version	Date	Description
1.0	23.6.2026	Initial Public Report Version



Legal Notice and Permitted Use

Scope: This report is issued by Tropic Square to inform the public about a security evaluation of a Tropic Square product, the general nature and potential impact of the findings, and any remediation or mitigation measures taken, where applicable. This notice applies to all information contained in or accompanying the report.

Permitted Use and Legal Compliance: This report, including any technical information it contains, is provided solely for security awareness, defensive validation, remediation, and authorized security testing. Readers are responsible for complying with all applicable laws, regulations, and contractual obligations, and must not use this report for any unauthorized, harmful, or unlawful purpose.

No Warranty and Liability Limitation: This report is provided on an “as is” basis. Tropic Square makes reasonable efforts to provide accurate and useful information but does not warrant that the report is complete, error-free, applicable to every environment, or sufficient to eliminate all security risks. This report also does not constitute a warranty regarding any product’s suitability for a particular purpose. To the maximum extent permitted by applicable law, Tropic Square shall not be liable for any damages, losses, claims, or disruptions arising from the use, misuse, inability to use, or reliance on this report.

Repository Consistency Disclaimer: Where this report refers to public repositories or published materials, readers should note that such materials may not always identically correspond to a specific product version, configuration, or release. Differences may arise due to publication timing, implementation details, hardware integration, third-party rights, or other technical or legal limitations. Any use, copying, modification, or redistribution of materials from Tropic Square repositories is governed by the applicable license terms in those repositories.

Intellectual Property: Nothing in this report grants any intellectual property right, except for a limited, non-exclusive, revocable permission to read, reference, quote, and share the report for lawful security awareness, risk assessment, and remediation purposes. Any sharing must preserve attribution to Tropic Square, must not be misleading, and must not present outdated versions as current. Redistribution is permitted only in complete and unmodified form, except for clearly identified quotations, references, or summaries that do not misrepresent the report.

Limited Disclosure of Detailed Findings: This public report may be based on one or more underlying evaluation reports. For security reasons, Tropic Square may omit,



redact, delay, generalize, or withhold detailed findings or other sensitive technical information where disclosure could increase risk to users, customers, partners, or other affected parties. Such information may be protected by confidentiality measures to reduce the risk of misuse.

Authorized Testing: Tropic Square encourages good-faith security research conducted in accordance with its Coordinated Vulnerability Disclosure Policy (CVD). Any reported weakness or vulnerability will be handled in accordance with that policy and may, where appropriate, form the basis of further public reports. Publication of this report does not authorize testing or other security research outside the scope of that policy.

Researcher Acknowledgement: Tropic Square may acknowledge researchers who contributed to relevant findings, subject to their consent. Any such acknowledgement does not imply employment, agency, endorsement, partnership, authorization to test outside the CVD Policy, or approval of any conduct not expressly permitted by that policy.

Updates: Tropic Square may update this report from time to time, and readers should verify that they are relying on the latest available version. Where this report conflicts with later Tropic Square advisories, errata, release notes, product documentation, or direct security communications, the later or more specific communication controls unless stated otherwise.

Contact: Questions regarding this report should be submitted through Tropic Square's official security contact channel: security@tropicsquare.com. Suspected vulnerabilities should be reported in accordance with the CVD Policy: <https://www.tropicsquare.com/cvd>.



Tropic Square Response to the Evaluation Report

The partner evaluation report has identified several weaknesses and bugs in preproduction open samples of TROPIC01. This section serves to summarize the findings and provide Tropic Square's response, along with the mitigations already implemented and those planned for these findings. We invite the reader to first read the partner evaluation report which can be found in the next section of this document.

Supply voltage monitor not enabled after power-up (Executive Summary, Sec. 3.1.2)

The supply voltage monitor (low-voltage alarm) is not enabled by the RISC-V ROM bootloader after power-up, due to a ROM bootloader bug present up to silicon revision C (ACAB).

The bug has been fixed for the next silicon revisions. Additionally, an updated Mutable FW version has been released (v0.3.1) that enables the supply voltage monitor at the earliest possible stage of the boot process - upon entrance to Idle mode [1]. See the errata for this issue in [2].

Faults injected without triggering sensors

For all three evaluated fault injection techniques — voltage, electromagnetic, and laser — faults might sometimes be injected without triggering any of the hardware sensors.

Voltage fault injection. (Sec. 3.1.2) Even with the voltage monitor and glitch detector active, a small number of faults (six out of approximately one million injected glitches) were injected that set only the supply voltage monitor alarm (which is not enabled during Start-up mode for silicon revision C).

Electromagnetic fault injection. (Sec. 3.2.2) EM pulses can be injected without triggering any alarm at low probability.

Laser fault injection. (Sec. 3.3.2) Several regions were identified where faults can be reliably injected without triggering any sensor, both in the standard-cell logic area that contains the laser sensors and in the memory arrays.

Tropic Square's reaction. Hardware sensors cannot provide complete coverage against fault injection attacks. Following the principle of defense in depth, the RISC-V and SPECT ROM bootloaders have been hardened against fault injection at the software level to complement the sensors in the next silicon revision. Moreover, a boot-hold feature has been added to the next silicon revision, which reacts to multiple alarm triggers between device power cycles and rate-limits the attacker's ability to perform consecutive faults.



We would also like to note that these findings only demonstrate undetected faults that alter the chip's behavior. A change in behavior does not necessarily mean it can be exploited to mount an attack. Fault injection using different techniques, equipment or environmental conditions may yield different results.

Considering the fault parameters evaluated in this report, the combination of the supply voltage monitor and glitch detector provides a strong defense against voltage fault injection — for the C silicon revision after Start-up mode, and for future silicon revisions during the whole device operation. No undetected faults have been observed in a scenario with both the supply voltage monitor and glitch detector enabled.

Electromagnetic and Laser fault injections are attack techniques requiring specialized equipment and preparation. In future silicon revisions, the FW-level fault injection hardening together with a properly configured boot-hold feature will significantly increase the attacker's effort to successfully inject faults leading to exploitable behavior compared to the current silicon revision C.

SPECT EdDSA signature-verification bypass (Sec. 4)

Logically skipping a single `JMP set_res_word` instruction at the end of the SPECT EdDSA verify implementation causes an invalid signature to be reported as valid. COSIC confirmed this vulnerability in practice via laser fault injection. For the next silicon revision, the SPECT boot code has been fixed to prevent this specific signature-verification bypass. [4]

Additionally, Tropic Square has developed a fault emulation tool that applies a fault model derived from the real-world fault testing done by COSIC. Using this tool, many other theoretically possible fault paths leading to an EdDSA verify bypass have been identified and fixed for the next silicon revision. SPECT fault emulation has been integrated as an automated testing tool into the development process.

Adversarially crafted inputs for SPECT EdDSA signature-verification bypass (Sec. 5.2)

Instead of solely focusing on the critical checks at the end of the algorithm it is also possible to purposefully craft a malicious input and inject a fault during the EdDSA computation [4]. Several of such paths have been identified by COSIC.

All identified paths have been fixed in the SPECT boot code for the next silicon revision. These scenarios have also been added to the fault emulation tool, making them part of the automated test pipeline used during development.



Summary

Tropic Square acknowledges that hardware security sensors cannot detect all possible types of fault injection techniques, and that they are therefore only one layer in a defense-in-depth strategy. All exploitable attack paths and bugs identified by COSIC have been mitigated partially in the current silicon revision through a firmware update, and fully in the next silicon revision through measures in the RISC-V and SPECT ROM bootloader code.



The Partner Evaluation Report Follows

KU Leuven, Computer Security and Industrial Cryptography (COSIC)

Cooperation between COSIC and Tropic Square

Assessing the feasibility of fault injection attacks on TROPIC01

– Public report –

Authors: Dr. Lennert Wouters, Dr. Benedikt Gierlichs

Leuven, May 2026



1 Executive Summary

At the request of Tropic Square, the COSIC research group at KU Leuven in Belgium performed a scope-limited evaluation of the TROPIC01 secure element. This report provides technical details of the evaluation and its results.

As part of this evaluation, COSIC was asked to validate some of the integrated security sensors that are designed to detect specific means of fault injection. More precisely, COSIC studied the following sensors:

- supply voltage monitor
- voltage glitch detector
- EM pulse detector
- laser detector

In general, the sensors were active and detected the fault injection mechanisms they are designed to detect. However, the supply voltage monitor was observed to be misconfigured after power-up. This issue is addressed in the next silicon revision of TROPIC01. Additionally, in several cases, faults were injected without triggering any security sensors.

COSIC was further asked to verify if the discovered issues can be exploited to mount a successful attack. To that end COSIC demonstrated a successful laser fault injection attack against the signature verification engine which is used to protect software updates and secure boot.

All evaluation results were reported to Tropic Square. Tropic Square decides on a case-by-case basis if and how issues are addressed.

1.1 Disclaimer

The information in this report is provided “as is” without warranty of any kind, express or implied, including but not limited to warranties of accuracy, completeness, fitness for a particular purpose, or non-infringement. Use of this information is at your sole risk.

2 Introduction

The Tropic Square TROPIC01 secure element integrates a Security Center. The Security Center is, among other things, responsible for detecting anomalies and handling 35 distinct alarms. Some of these alarms are generated by dedicated hardware sensors that are designed to detect physical attacks. While these sensors are meant to detect physical attacks, they are unlikely to provide complete coverage.

Therefore, the primary goal of this evaluation was to establish the functionality and efficacy of the different security sensors in TROPIC01. This involves performing controlled fault injection experiments on TROPIC01, including voltage, electromagnetic, and laser fault injection.

Ultimately, our evaluation led to the identification of two confirmed vulnerabilities. First, the supply voltage monitor was not enabled by the RISC-V

ROM bootloader, this enabled successful voltage fault injection. Second, a realistic laser fault injection attack path was identified targeting the TROPIC01 SPECT Edwards-curve Digital Signature Algorithm (EdDSA) [3] verify implementation. This component is security-critical, as it verifies firmware at boot time, and during firmware updates.

Additionally, we built experimentally derived fault models for the SPECT engine. These fault models were used to simulate the effects of fault injection in the SPECT EdDSA implementation. The simulation results were used by Tropic Square to produce a new implementation hardened against fault injection.

The identified issues are addressed through revisions of the RISC-V and SPECT ROM bootloaders in the next TROPIC01 silicon revision.

3 Exploration of multiple fault injection techniques

All of the experiments covered in this report were performed using a Tropic Square TS0102 base board and multiple TS02 TROPIC01 target boards. For the purpose of this evaluation, Tropic Square provided open samples that allow execution of arbitrary programs, as well as all required documentation and technical support. Confidential documentation of integrated 3rd party IP was not available during the evaluation.

For the experiments in this section, a simple RISC-V test program was used. The program increments a counter within nested loops, and a general purpose output pin is used as a trigger signal for fault injection synchronization. Additionally, multiple exception handlers are implemented to help detect anomalous behavior. All the sensors are enabled through the debugging interface. After each fault injection attempt, the output of the target program and the alarm status registers are retrieved and stored for analysis.

This experimental setup favors the evaluator: the device under test provides a trigger signal, extensive feedback is available to the evaluator, and the code is not hardened against fault injection.

3.1 Voltage fault injection

TROPIC01 includes a supply voltage monitor and a power supply glitch detector. The voltage monitor is designed to detect whether VCC is within a predefined operating voltage range. If VCC is outside this operating range, an alarm should be raised. The voltage monitor is not designed to detect fast transients caused by, e.g. voltage glitching. The glitch detector is designed to detect fast positive and negative transients.

Voltage Fault Injection (VFI) is commonly used to try and circumvent security features of integrated circuits. Low-cost tools to perform VFI are commercially available. A wide range of adversaries use VFI as it is an accessible and effective technique. Therefore it is important to evaluate the susceptibility of TROPIC01 to VFI.

3.1.1 Experimental setup

We applied a commonly used VFI technique in which a voltage glitch is inserted in the supply voltage by quickly switching between a nominal voltage and a glitch voltage using an analog multiplexer [1, 2]. Specifically, a MAX4619 analog multiplexer and a two-channel programmable power supply were used to generate the nominal and glitch voltages. Glitch offset (time to wait after the trigger) and width were controlled through a ChipWhisperer Husky [7].

For the VFI experiments we modified the TS02 target boards such that power could be supplied directly through the multiplexer instead of the base board. Only the main 3.3 V supply is externally accessible; internal core voltages are not. Additionally, the nominal supply voltage was connected to the IO level shifters on the TS0102 base board. This ensured stable communication even at lower operating voltages.

3.1.2 Results

We started by performing an initial broad random parameter sweep to characterize the operating range of TROPIC01, and to establish whether or not we could trigger any of the sensors. This showed that both the glitch and supply voltage monitor sensors can be readily triggered. In later experiments we observed that the supply voltage monitor alarm was inactive unless manually enabled through the debug interface. This behavior was caused by a TROPIC01 ROM bootloader bug present up to silicon revision ACAB.

In further VFI experiments a fault is classified as undetected if no alarms other than the low-voltage alarm are set, this to quantify the impact of this ROM bootloader bug. Figure 1 shows a plot representing one VFI experiment consisting of approximately one million injected voltage glitches. The represented dataset contains six of these undetected faults, all of these did have voltage monitor alarms set.

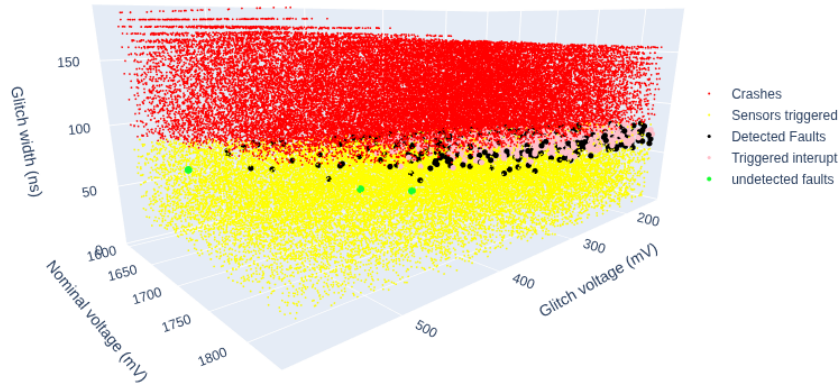


Figure 1: A graphical representation of one VFI experiment. Note that a fault is only classified as undetected if no alarms besides the low voltage alarm are set. We did this to demonstrate the impact of the low voltage detector not being enabled in TROPIC01 boot ROM revisions up to revision ACAB.

Based on these experiments we concluded that the combination of the voltage monitor and the glitch detector detects all of the voltage glitches we injected. Even though better results may be achievable with different parameters or VFI techniques (see Section 3.4), this attack vector appears less effective than other fault injection methods.

3.2 Electromagnetic fault injection

Electromagnetic fault injection is another commonly used fault injection technique that allows targeting of a more localized area of the device under test. Commercial EMFI tools are available and are generally more expensive than VFI tools. However, several (open-source) options exist that are within reach of hobbyists. TROPIC01 includes sensors designed to detect EM pulses, our goal was to assess their effectiveness.

3.2.1 Experimental setup

The EMFI experiments were performed using a NewAE ChipShouter (1mm CCW and CW probes) and a Langer BPS 202 supply with an ICI HH500-15 L-EFT probe. The ChipShouter was used from the top side of an unmodified TROPIC01 package, while the Langer system was used for backside EMFI on a decapsulated TROPIC01 (see Figure 2).

Backside decapsulation of TROPIC01, required for the EMFI and LFI experiments, was performed manually. We used an electrical grinding pen to remove the majority of the epoxy molding around the copper paddle, which was then detached by prying with a scalpel. The conductive epoxy between the paddle and die was softened with acetone before being scraped away.

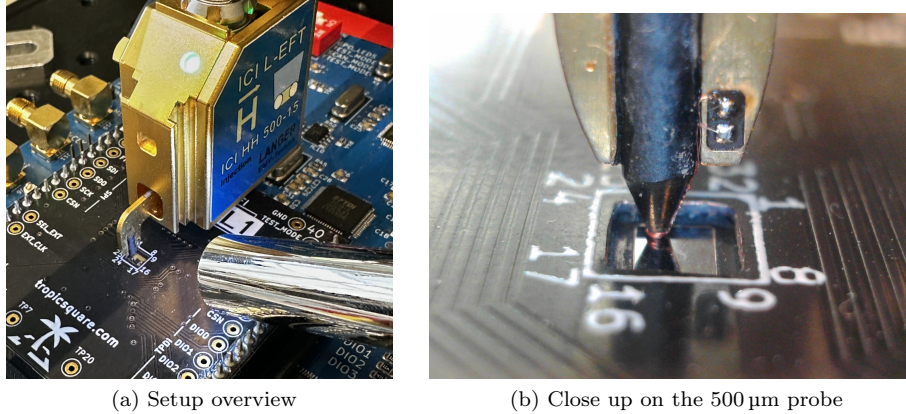


Figure 2: Using a small EMFI probe on the backside of the TROPIC01 die.

3.2.2 Results

EMFI experiments over wide parameter ranges showed that while many EM pulses are detected, faults can still be injected without triggering any alarm.

ChipShouter front side EMFI: We found that, with low probability, undetected faults can be injected in a specific corner of the TROPIC01 package using the ChipShouter and its included 1 mm CW probe.

Langer backside EMFI: EMFI from the back side of the TROPIC01 using the ICI HH500-15 L-EFT probe was more effective and repeatable than the ChipShouter applied from the front side. However, using the ICI HH500-15 L-EFT probe from the front side (on top of the chip package) did not produce observable faults. Figure 3 shows the result of one EMFI experiment that provides an indication for the sensitivity of the EM sensors.

Further work is required to determine whether the observed differences are due to probe locality, pulse characteristics (e.g., rise time), probe distance, or injection side (front vs. back).

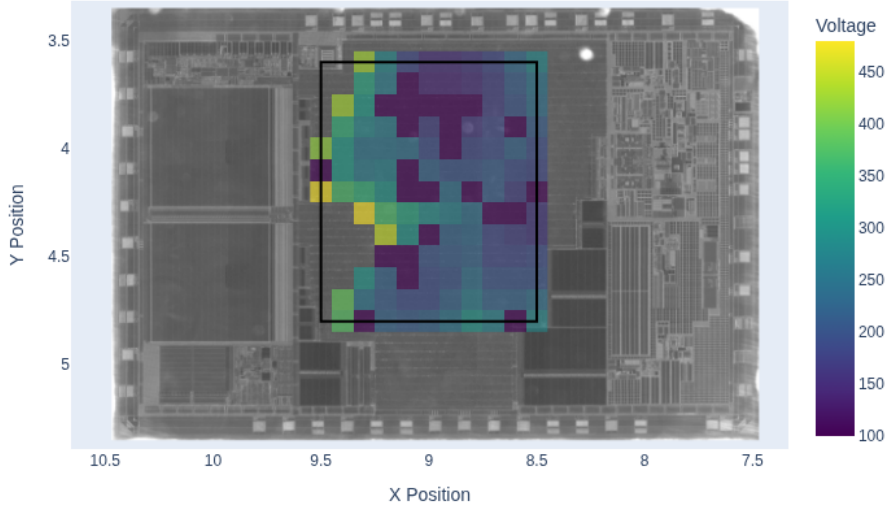


Figure 3: Result of an EMFI experiment using the Langer ICI HH500-15 L-EFT approaching from the backside of the die. This plot indicates the minimum pulse voltage that was required to trigger the EMFI alarm per probe location within the scanned area (indicated by the black rectangle). Locations within the rectangle that do not have a color assigned did not result in any EMFI alarm.

3.3 Laser fault injection

Laser Fault Injection (LFI) is the most common optical fault injection technique [9]. It is considered a semi-invasive fault injection technique and it is typically chosen for its superior locality. The standard cell logic area of the TROPIC01 secure element includes sensors to detect LFI attempts. This section assesses the functionality of these sensors, in Section 4 we use these results to construct a realistic attack.

3.3.1 Experimental setup

We used a custom LFI setup with an Alphanov 1064nm laser source focused through the silicon substrate using a 50x Mitutoyo M Plan APO NIR objective. Figure 4 shows a through-substrate infrared image of the TROPIC01 secure element acquired using the same setup. An annotated version of this image (provided by Tropic Square) and photon emission experiments were used to guide our parameter search. Laser output power and pulse width were varied during the experiments.

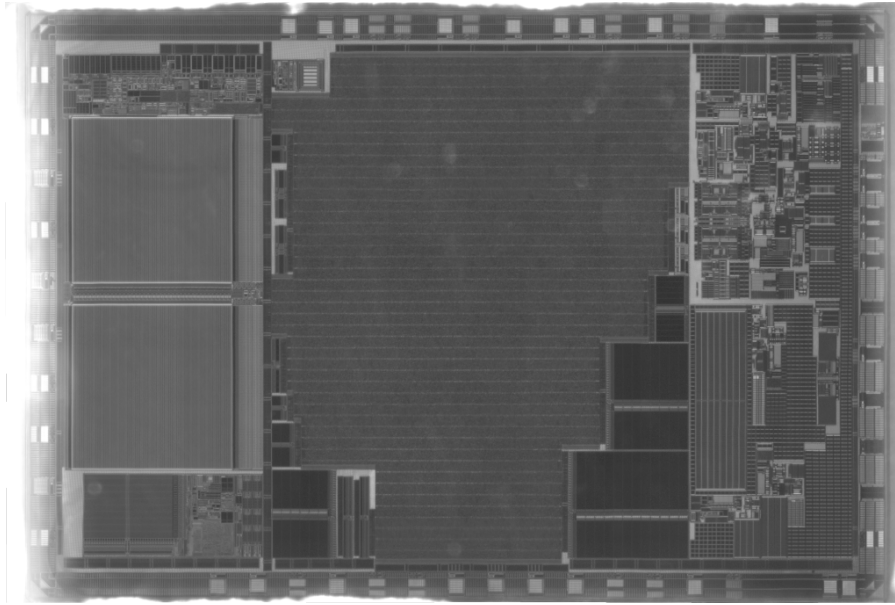


Figure 4: A through-substrate infrared image of the TROPIC01 secure element.

3.3.2 Results

The laser sensors detected many injected laser pulses. However, we were also able to identify several regions where we could reliably inject faults without triggering any sensor. This was the case both in the standard cell logic area that contains the laser sensors, and in the memory arrays. Exact locations and parameters are not disclosed in this report to avoid providing information to potential adversaries.

3.3.3 Sensor characterization

After establishing that LFI can be used to inject faults, we performed extensive laser sensor characterization experiments. One experiment involved creating detailed maps (see Figure 5) showing, per position, the minimum energy required (in our LFI setup) to inject a fault, as well as the minimum energy required to trigger the sensor. As expected, the energy required to trigger an alarm decreases as the laser spot approaches the sensor location.

These experiments provide Tropic Square insight into operating ranges, effective coverage, and limitations of the sensor IP used in TROPIC01, and support the development of improved future designs.

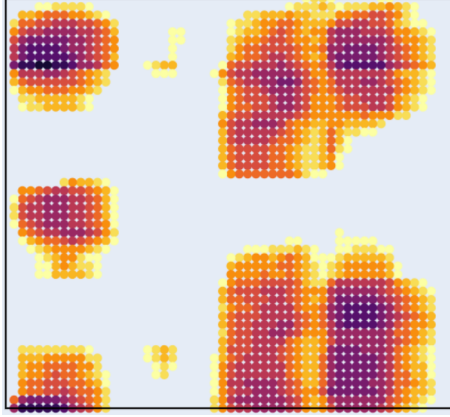


Figure 5: A laser sensor sensitivity map. This map is redacted to avoid disclosing the physical location and energy required to trigger the sensor. Each colored dot indicates the minimum laser source output power required to trigger a laser sensor in that specific location. Darker colors indicate the regions where the laser sensors are the most sensitive. In other words, a darker dot indicates that a lower laser output power is required to trigger the laser sensor. Locations that do not have a colored dot did not result in any laser alarm within the pre-defined laser output power range used during the characterization.

3.4 Scope and limitations

Our evaluation focused on fault injection; other attack vectors and system-level responses to these alarms were not evaluated.

The fault injection parameter space is large, so we were unable to test all possible (variations of) techniques and parameter combinations in the allotted time. Observed results may differ when using alternative techniques (e.g. crow-bar VFI [6], arbitrary waveform VFI [4], Body Bias Injection [5]), or when such techniques are applied under different environmental conditions.

4 Targeting SPECT with fault injection

The experiments covered in Section 3 demonstrate that it is possible to inject faults. Our next goal was to assess if these faults can be used in a realistic scenario rather than a dummy program.

Based on discussions with Tropic Square we decided that SPECT is a particularly interesting target from an adversary’s perspective. SPECT is a hardware co-processor for Elliptic Curve Cryptography (ECC) operations. Among other tasks, SPECT verifies an EdDSA signature over new firmware updates.

The following enumeration describes the EdDSA signature verification implementation that is used in SPECT, using the same notation as what is used

in the implementation. Where a signature (R, s) , over a message (M) is verified with the public key (A) . The provided signature is considered valid if the final check passes.

1. $Q_1 = s.B$
2. $E = SHA512(R||A||M) \bmod q$
3. $Q_2 = E.A$
4. $Q = Q_1 - Q_2$
5. $R == Q$

A cursory analysis of the initial SPECT EdDSA implementation indicated that skipping a single `JMP` instruction during the final $R == Q$ check could bypass signature verification. As shown in the following listing, for an invalid signature the SPECT engine will end up executing the code under the `eddsa_verify_fail` label ¹. If the `JMP set_res_word` instruction is logically skipped then the code under the `eddsa_verify_success` label will be executed. This causes the code to load the success magic value in `eddsa_verify_output_result`, thereby indicating successful signature verification.

Listing 1: The end of the EdDSA verify implementation.

```

CMPI      r31 , 0
XOR       r2 , r23 , r8
BRZ       eddsa_verify_success

eddsa_verify_fail:
MOVI      r2 , 1
ST        r2 , eddsa_verify_output_result
JMP       set_res_word

eddsa_verify_success:
MOVI      r2 , 0xBA
ROL8      r2 , r2
ORI       r2 , r2 , 0x11
ROL8      r2 , r2
ORI       r2 , r2 , 0xFA
ROL8      r2 , r2
ORI       r2 , r2 , 0xDE
ROL8      r2 , r2
ORI       r2 , r2 , 0x0B
ROL8      r2 , r2
ORI       r2 , r2 , 0x5E
ROL8      r2 , r2
ORI       r2 , r2 , 0x55
ROL8      r2 , r2
ORI       r2 , r2 , 0xED
ST        r2 , eddsa_verify_output_result
JMP       set_res_word

```

¹eddsa_verify.s, lines 134–156

To assess the feasibility of this attack strategy we first targeted a dummy SPECT program consisting of a sequence of `ADDI` and `JMP` instructions. This confirmed that we could logically skip a `JMP` instruction via LFI. Subsequently, we also performed this attack on the actual SPECT EdDSA implementation, confirming that this vulnerability is exploitable. Section 5 evaluates the broader attack surface of the EdDSA implementation and proposes mitigations.

5 Evaluation of the EdDSA implementation in SPECT

After demonstrating a successful attack on the EdDSA implementation, our next objective was to suggest improvements. To that end we performed a manual review of the code and conducted additional automated fault simulation experiments. The results of this evaluation were used by Tropic Square to harden their SPECT EdDSA implementation against fault injection attacks to further complement the sensors.

5.1 Manual and simulated analysis

We started with a manual evaluation of the implementation assuming attacker-controlled message and signature inputs, as would be the case for a firmware update. Schneider et al. provide an overview of fault injection strategies for ECC-based signature verification protocols [8]. Instead of solely focusing on the critical checks at the end of the algorithm it is also possible to purposefully craft the input and inject a fault during the computation. We identified several potential attack paths that require either standard or adversarially crafted inputs. See Section 5.2 for an example of an attack scenario that required adversarially crafted inputs.

However, we also determined that exhaustive manual evaluation of the EdDSA code would be infeasible. Additionally, we find that during manual evaluation it is easy to get stuck into an "instruction skip" mental model, which may be incomplete or unrealistic.

Therefore, we performed several practical LFI experiments to build realistic fault models for the SPECT engine. These fault models were used to perform a more rigorous evaluation of the EdDSA implementation through fault simulation. We developed a proof-of-concept fault simulator for SPECT as a Python wrapper for the SPECT Instruction Set Simulator (ISS) [10]. These simulation results informed the development of an implementation hardened against fault injection by Tropic Square. Additionally, Tropic Square integrated the fault simulation approach into their own validation pipeline.

Manual analysis identified 9 unique single-fault attack paths and 14 input test vectors that were used as inputs for fault simulation. The fault simulation identified more than 70 single-fault attack paths across these test vectors.

5.2 Adversarially crafted input

In Section 4.3.3 of their paper Schneider et al. provide several potential attack paths that can affect EdDSA implementations [8]. The applicability of these

scenarios strongly depend on exact implementation details and considered fault models.

The following attack scenario is not specifically mentioned in the work by Schneider et al. , but it is one example of an attack scenario that affects the original SPECT EdDSA implementation. Note in Listing 2 that the SHA512 operation is implemented by the `HASH_IT` (to reset the state) and `HASH` instructions. If the `HASH` instruction is skipped (or modified into a `HASH_IT` instruction), we would end up with `r28` being unaffected instead of containing the lower half of the hash digest. In the studied implementation that means that `r28` will contain s , resulting in the check $R == s.B - (s \bmod q).A$, this is an exploitable scenario as an adversary can craft inputs that satisfy this check.

Listing 2: The start of the SHA512 calculation.

```

;
;  E = SHA512(ENC(R) || ENC(A) || M) mod q
;
LD      r25, eddsa_verify_input_message
LD      r26, eddsa_verify_input_pubkey
LD      r27, eddsa_verify_input_R

SWE     r21, r25
SWE     r22, r26
SWE     r23, r27

; padding
MOVI    r20, 1
ROR     r20, r20
ORI     r20, r20, 0x300

HASH_IT
HASH    r28, r20

```

5.3 Mitigations

The manual and the automated analysis resulted in a set of recommendations to harden the SPECT EdDSA implementation against faults. These recommendations were implemented by Tropic Square to produce a hardened SPECT EdDSA implementation. This new implementation was verified using the fault injection simulator and will be included in the next TROPIC01 silicon revision.

6 Conclusion

Defense in depth is essential for security critical systems, including secure elements. Security sensors designed to detect physical attacks provide a strong defense layer but cannot prevent all attacks. Multiple independent defense layers are therefore required. Our evaluation of the TROPIC01 secure element demonstrates that effective faults can be injected without triggering security sensors, and that such faults can enable realistic attacks.

Based on our results Tropic Square decided to harden both the RISC-V and SPECT ROM bootloaders, further improving the resilience of TROPIC01 against physical attacks. These updates will be included in future TROPIC01 silicon revisions.

References

- [1] Chris Gerlinsky. Breaking code read protection on the NXP LPC-family microcontrollers. REcon, 2017. https://recon.cx/2017/brussels/resources/slides/RECON-BRX-2017-Breaking_CRP_on_NXP_LPC_Microcontrollers_slides.pdf.
- [2] Michael Hutter, Jörn-Marc Schmidt, and Thomas Plos. Contact-based fault injections and power analysis on RFID tags. In *19th European Conference on Circuit Theory and Design, ECCTD 2009, Antalya, Turkey, August 23-27, 2009*, pages 409–412. IEEE, 2009.
- [3] Simon Josefsson and Ilari Liusvaara. Edwards-Curve Digital Signature Algorithm (EdDSA). RFC 8032, January 2017.
- [4] Timo Kasper, David F. Oswald, and Christof Paar. A versatile framework for implementation attacks on cryptographic rfids and embedded devices. *Trans. Comput. Sci.*, 10:100–130, 2010.
- [5] Philippe Maurine. Techniques for EM fault injection: Equipments and experimental results. In *2012 Workshop on Fault Diagnosis and Tolerance in Cryptography*, pages 3–4, 2012.
- [6] Colin O’Flynn. Fault injection using crowbars on embedded systems. Cryptology ePrint Archive, Report 2016/810, 2016.
- [7] Colin O’Flynn and Zhizhang (David) Chen. ChipWhisperer: An open-source platform for hardware embedded security research. pages 243–260, 2014.
- [8] Kevin Schneider, Lukas Auer, and Alexander Wagner. Fault attacks on ECC signature verification. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2025(4):1010–1052, Sep. 2025.
- [9] Sergei P. Skorobogatov and Ross J. Anderson. Optical fault induction attacks. pages 2–12, 2003.
- [10] Tropic Square. ts-spect-compiler: Spect compiler. <https://github.com/tropicsquare/ts-spect-compiler>, 2024. Accessed: 2026-05-21.

– This line marks the end of the report.



The End of Partner Evaluation Report



References

- [1] ODD_TR01_datasheet_vA_10
https://github.com/tropicsquare/tropic01/blob/main/doc/datasheet/ODD_TR01_datasheet_vA_10.pdf
- [2] OI_TR01_ERR_2026060200: Voltage monitor not enabled during Immutable FW operation
<https://tropic-ops.atlassian.net/wiki/external/MjU4ZDY3NjQ2MzAzNDM0NDhhYmFjMWIzN2Y5ZWY0MmQ>
- [3] Common Criteria and Open-Hardware Evaluation Methodology in the ORSHIN Project
https://github.com/ORSHIN/public-assets/blob/public/risk-assessment/CC_EVALUATION.md
- [4] ODR_TR01_SA_2026012900: Potential Bypass of Firmware Verification by Laser Fault Injection.
https://github.com/tropicsquare/tropic01/blob/main/doc/security_advisories/ODR_TR01_SA_2026012900_1v0.pdf